Table 1: Comparison of change detectors. Complexity — runtime complexity per new observation, ARL / MTD — type of known results, domain — data types, t — total number of observations, d — dimensionality (for Euclidean spaces), k — parameter, W — window length / block size, N — number of windows.

Algorithm	Complexity	ARL / MTD	Domain
ADWINK	$\mathcal{O}(dk \log W)$	empirical	$\mathbb{R}^d$
WATCH	unknown[a]	empirical	$\mathbb{R}^d$
Scan B	$\mathcal{O}(NW^2)$	analytical	topological
NEWMA	$\mathcal{O}(md)$[b]	analytical	$\mathbb{R}^d$
D3	$\mathcal{O}(W^3)$[c]	none	$\mathbb{R}^d$
IBDD	$\mathcal{O}(pq)$[d]	none	$\mathbb{R}^d$
MMDEW	$\mathcal{O}(\log^2 t)$	empirical	topological

$x_t \in \mathcal{X}$. Each x_t is generated independently following some distribution $D_t \in \mathcal{M}_1^+(\mathcal{X})$. If there exists t^* such that for $i < t^*$ and $j \geq t^*$ we have $D_i \neq D_j$, then t^* is a change point, and our task is to detect it; in practice, a D_t typically generates a range of i.i.d. observations. We note that these definitions place few assumptions on the type of data, that is, we only require the data to reside in a topological space.

Kernel mean embedding. Let $\mathcal{H}$ be a reproducing kernel Hilbert space (RKHS) on $\mathcal{X}$, which means that the linear evaluation functional $\delta_x : \mathcal{H} \to \mathbb{R}$ defined by $\delta_x(f) = f(x)$ is bounded for all $x \in \mathcal{X}$ and $f \in \mathcal{H}$. By the Riesz representation theorem (Reed & Simon, 1972), there exists for each $x \in \mathcal{X}$ a unique vector $\phi(x) \in \mathcal{H}$ such that for every $f \in \mathcal{H}$ it holds that $f(x) = \delta_x(f) = \langle f, \phi(x) \rangle$. The function $\phi(x)$ is the reproducing kernel for x and also called feature map; it has the canonical form $x \mapsto k(\cdot, x)$, with the function $k : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ the reproducing kernel associated to $\mathcal{H}$. With this kernel, it holds that $k(x_1, x_2) = \langle \phi(x_1), \phi(x_2) \rangle = \langle k(\cdot, x_1), k(\cdot, x_2) \rangle$ for all $x_1, x_2 \in \mathcal{X}$ (Steinwart & Christmann, 2008). The mean embedding of a probability measure $\mathbb{P} \in \mathcal{M}_1^+(\mathcal{X})$ is the element $\mu(\mathbb{P}) \in \mathcal{H}$ such that $\mathbb{E}_{X \sim \mathbb{P}}[f(X)] = \langle f, \mu(\mathbb{P}) \rangle$ for all $f \in \mathcal{H}$. The mean embedding $\mu(\mathbb{P})$ exists if k is measurable and bounded (Sriperumbudur et al., 2010, Prop. 2), which we assume throughout the article.

Maximum mean discrepancy. MMD is defined by $\mathrm{MMD}(\mathbb{P}, \mathbb{Q}) = \|\mu(\mathbb{P}) - \mu(\mathbb{Q})\|$, where $\mu(\mathbb{P}), \mu(\mathbb{Q}) \in \mathcal{H}$ are the mean embeddings of $\mathbb{P}, \mathbb{Q} \in \mathcal{M}_1^+(\mathcal{X})$, respectively.

Let $X \sim \mathbb{P}$, $Y \sim \mathbb{Q}$ and X', Y' independent copies of X, Y, respectively. The squared population MMD (Gretton et al., 2012, Lemma 6) then takes the form

$$\mathrm{MMD}^2(\mathbb{P}, \mathbb{Q}) = \mathbb{E}[k(X, X')] + \mathbb{E}[k(Y, Y')] - 2\mathbb{E}[k(X, Y)],$$

where the expectations are taken w.r.t. to all sources of randomness. For observations $\hat{\mathbb{P}}_m = \{x_1, \ldots, x_m\} \overset{\text{i.i.d.}}{\sim} \mathbb{P}$ and $\hat{\mathbb{Q}}_n = \{y_1, \ldots, y_n\} \overset{\text{i.i.d.}}{\sim} \mathbb{Q}$, a biased estimator is obtained by replacing the population means with their empirical counterparts

$$\mathrm{MMD}^2\left(\hat{\mathbb{P}}_m, \hat{\mathbb{Q}}_n\right) = \frac{1}{m^2} \sum_{i,j=1}^{m} k(x_i, x_j) + \frac{1}{n^2} \sum_{i,j=1}^{n} k(y_i, y_j) - \frac{2}{mn} \sum_{i,j=1}^{m,n} k(x_i, y_j). \tag{1}$$

The runtime complexity of (1) is in $\mathcal{O}(m^2 + n^2)$. We will base our proposed approximation on (1).

Two-sample testing. To decide whether the value of $\mathrm{MMD}\left(\hat{\mathbb{P}}_m, \hat{\mathbb{Q}}_n\right)$ indicates a significant difference between $\mathbb{P}$ and $\mathbb{Q}$, one tests the null hypothesis $H_0 : \mathbb{P} = \mathbb{Q}$ versus its alternative $H_1 : \mathbb{P} \neq \mathbb{Q}$ by defining an acceptance region for a given level $\alpha \in (0, 1)$, which takes the form $\mathrm{MMD}\left(\hat{\mathbb{P}}_m, \hat{\mathbb{Q}}_n\right) < \epsilon_\alpha$. One rejects H_0 if the test statistic exceeds the threshold. The level α is a bound for the probability that the tests rejects H_0 incorrectly (Casella & Berger, 1990). Assuming that k is nonnegative and bounded by $K > 0$, that is, $0 \leq k(x, y) \leq K$ for all $x, y \in \mathcal{X}$, Gretton et al. (2012, Corollary 9) provides the distribution-free threshold

ϵ_α for the case that both samples $\hat{\mathbb{P}}_m$ and $\hat{\mathbb{Q}}_m$ have the same size ($m = n$) as

$$\mathrm{MMD}\left(\hat{\mathbb{P}}_m, \hat{\mathbb{Q}}_m\right) < \sqrt{\frac{2K}{m}}\left(1 + \sqrt{2\log\frac{1}{\alpha}}\right). \tag{2}$$

Computing (2) costs $\mathcal{O}(1)$. As the change detection setting requires the case that $m \neq n$, we extend their threshold accordingly in what follows.

4 Our proposed algorithm

We introduce MMDEW in three steps. We first extend the threshold for the MMD two-sample test, (2), to samples of unequal sizes (Section 4.1). We then introduce our data structure that enables the efficient computation of MMD on data streams (Section 4.2). Last, we describe the complete algorithm in Section 4.3.

4.1 Threshold for the hypothesis test

Given a sequence of observations $\{x_1, \ldots, x_t\}$ up until time t our goal is to test the null hypothesis $\mathbb{P} = \mathbb{Q}$ for any two neighboring windows $X \cdot Y = \{x_1, \ldots, x_i\} \cdot \{x_{i+1}, \ldots, x_t\}$, with $i = 1, \ldots, t-1$. Our following proposition extends Gretton et al. (2012, Theorem 8), which considers the case $m = n$, giving the distribution-free acceptance region for $m \neq n$ (corresponding to the setting that one generally encounters in change detection). The proof is deferred to Appendix A.1.

Proposition 1. *Let $\mathbb{P}, \mathbb{Q} \in \mathcal{M}_1^+(\mathcal{X})$, $\hat{\mathbb{P}}_m = \{x_1, \ldots, x_m\} \overset{i.i.d.}{\sim} \mathbb{P}$, $\hat{\mathbb{Q}}_n = \{y_1, \ldots, y_n\} \overset{i.i.d.}{\sim} \mathbb{Q}$. Assume that $0 \leq k(x, y) \leq K$ for all $x, y \in \mathcal{X}$ and $t > 0$. Then a hypothesis test of level at most $\alpha > 0$ for $\mathbb{P} = \mathbb{Q}$ has the acceptance region*

$$\mathrm{MMD}\left(\hat{\mathbb{P}}_m, \hat{\mathbb{Q}}_n\right) < \sqrt{\frac{K}{m} + \frac{K}{n}}\left(1 + \sqrt{2\log\alpha^{-1}}\right) =: \epsilon_\alpha.$$

Note that, when considering multiple possible change points, one needs to account for multiple testing in order to achieve an overall level of size α. For example, one may adjust ϵ_α through Bonferroni correction ($\epsilon'_\alpha = \epsilon_\alpha/(t-1)$) by dividing by the total number of tests.

To perform change detection with MMD, it is natural to consider the stopping time

$$T = \inf\left\{t : \max_{n=1,\ldots,t-1}\left[\mathrm{MMD}\left(\mathbb{P}_m, \mathbb{Q}_n\right) \geq \epsilon_\alpha\right] = 1\right\}, \tag{3}$$

for $m = t - n$, the empirical measures $\mathbb{P}_m = \{x_1, \ldots, x_m\}$, $\mathbb{Q}_n = \{x_{m+1}, \ldots, x_t\}$, and the brackets equal to one if their argument is true and zero otherwise (Graham et al., 1994); we note that $\epsilon_\alpha := \epsilon_\alpha(m, n)$ depends on the respective sizes of the subsamples considered. In other words, a change is indicated by the first time any MMD estimated across all splits exceeds its threshold. However, due to the quadratic runtime requirements of MMD, the computation of (3) costs $\mathcal{O}(t^3)$ for each new observation.

We now introduce our novel data structure that allows considering multiple possible change points efficiently.

4.2 Proposed data structure

One common method to obtain a good runtime complexity in change detection algorithms is to slice the data into windows of exponentially increasing sizes (Bifet & Gavaldà, 2007). Recent observations are collected in smaller windows, and older observations are grouped into larger windows. This leads to a fine-grained change detection in the recent past and more coarse-grained change detection in the distant past.

Our new data structure adopts this concept and, at the same time, facilitates the computation of MMD. In what follows, we first describe the properties of the proposed data structure. Then, we show how to update the data structure and explain its use for change detection.

4.2.1 Properties

We use 2 as the basis for the exponential slicing. Then, after observing t elements, the number of windows stored in the data structure corresponds to the number of ones in the binary representation of t. We may thus index the windows as $B_l, \ldots, B_0$ (in decreasing order), with the largest position being $l = \lfloor \log_2 t \rfloor$. A window does not exist if the binary representation of t at this position is zero.

If it exists, a window $B_s = (\mathrm{X}_s, \mathrm{XX}_s, \mathrm{XY}_s)$ at position $s = 0, \ldots, l$ stores 2^s observations

$$\mathrm{X}_s = \left\{ x_1^{(s)}, \ldots, x_{2^s}^{(s)} \right\}, \tag{4}$$

together with the summaries

$$\mathrm{XX}_s = \sum_{i,j=1}^{2^s} k\left(x_i^{(s)}, x_j^{(s)}\right), \tag{5}$$

$$\mathrm{XY}_s = \left\{ \underbrace{\sum_{i=1}^{2^s} \sum_{j=1}^{2^{s+1}} k\left(x_i^{(s)}, x_j^{(s+1)}\right)}_{=:\mathrm{XY}_s^{s+1}}, \ldots, \underbrace{\sum_{i=1}^{2^s} \sum_{j=1}^{2^l} k\left(x_i^{(s)}, x_j^{(l)}\right)}_{=:\mathrm{XY}_s^{l}} \right\}, \tag{6}$$

where $\mathrm{XX}_s \in \mathbb{R}$ is the sum of the kernel k evaluated on all pairs of the window's own observations, and XY_s stores a list of sums of the kernel evaluated on the window's own observations and the observations in windows coming before it.[4] Storing a list enables the efficient merging of windows, elaborated in Lemma 2. The length of the list XY_s equals the number of windows having observations older than window B_s and is at most $\lfloor \log_2 t \rfloor$. We use XY_i^j to represent the entry in XY_i that refers to the window B_j. Specifically, in (6), XY_s^{s+1} stores the interaction of B_s with B_{s+1}; similarly, XY_s^l stores its interaction with B_l.

Remark 1. *Given a stream of data $x_1, x_2, \ldots, x_t$, (4) corresponds to the mapping $x_i^{(s)} = x_\ell$, with $\ell = \sum_{j=s+1}^{\lfloor \log t \rfloor} 2^j [B_j \text{ exists}] + i$, where the bracket is one if the argument is true and zero otherwise (using Iverson's convention; Graham et al. 1994). A bucket B_j exists if the j-th right-most digit in the binary expansion of t is 1.*

We summarize two of the main properties of the data structure as lemmas. Lemma 1 establishes that one can compute the value of MMD between two windows with constant complexity. The proof follows from comparing (5) and (6) with (1). Lemma 2 shows that windows can be merged with logarithmic runtime complexity. These results provide our first steps towards efficiently computing MMD in a data stream.

Lemma 1. *Let B_{s+1} and B_s be any two neighboring windows with elements $\mathrm{X}_{s+1} = \left\{ x_1^{(s+1)}, \ldots, x_{2^{s+1}}^{(s+1)} \right\}$ and $\mathrm{X}_s = \left\{ x_1^{(s)}, \ldots, x_{2^s}^{(s)} \right\}$, and sums as defined by (5) and (6), respectively. Then*

$$\mathrm{MMD}^2\left(\mathrm{X}_{s+1}, \mathrm{X}_s\right) = \frac{1}{(2^{s+1})^2} \mathrm{XX}_{s+1} + \frac{1}{(2^s)^2} \mathrm{XX}_s - \frac{2}{(2^{s+1})(2^s)} \mathrm{XY}_s^{s+1},$$

with a computational complexity of $\mathcal{O}(1)$.

Lemma 2. *Merging two windows B_{s+1} and B_s into a new window B', such that B' stores (4), (5), and (6) costs $\mathcal{O}(\log t)$.*

Besides showing the result, the proof of Lemma 2 illustrates the steps that allow merging windows efficiently.

[4] Note that the superscript (s) of the $x_i^{(s)}$-s indicates the corresponding window B_s.